# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

**ANSWER KEY & MARKING SCHEME**

## Ch 3 Test: Metals and Non-metals

*Chapter / Topic: Metals and Non-metals*

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### SECTION A

**Q1.** Which metal is a liquid at room temperature? **[1]**

**Answer:** (d) Mercury. Bromine is a liquid too, but it is a non-metal.

**Q2.** Which of the following can be beaten into thin sheets? **[1]**

**Answer:** (b) Zinc. Metals are malleable.

**Q3.** Which metal is stored under kerosene? **[1]**

**Answer:** (c) Sodium. It reacts so vigorously with air and moisture that it can catch fire.

**Q4.** A metal oxide that reacts with both acids and bases to form salt and water is called: **[1]**

**Answer:** (b) an amphoteric oxide, e.g. $Al_2O_3$, ZnO

**Q5.** Ionic compounds conduct electricity: **[1]**

**Answer:** (a). The ions are free to move only in the molten state or in solution.

**Q6.** **Assertion (A):** Aluminium is more reactive than iron, yet aluminium articles do not corrode easily. **Reason (R):** Aluminium forms a thin, protective… **[1]**

**Answer:** (a). The oxide layer stops further reaction. Anodising makes this layer thicker.

### SECTION B

**Q7.** Zinc gives hydrogen gas with dilute hydrochloric acid, but copper does not. Why? Write the balanced equation for the reaction of zinc. **[2]**

**Answer:** Copper is less reactive than hydrogen (it is below hydrogen in the reactivity series), so it cannot displace hydrogen from the acid. Zinc is more reactive than hydrogen: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

- Reason (reactivity series): 1
- Balanced equation: 1

**OR, answer:** Nitric acid is a strong oxidising agent. It oxidises the hydrogen produced to water and is itself reduced to oxides of nitrogen. Magnesium and manganese give hydrogen with very dilute $HNO_3$.

- Reason (oxidising agent): 1
- Mg and Mn: ½ + ½

**Q8.** What are amphoteric oxides? Give two examples. **[2]**

**Answer:** Metal oxides that react with both acids and bases to give salt and water. Examples: aluminium oxide ($Al_2O_3$) and zinc oxide (ZnO). For example, $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ and $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$.

- Definition: 1
- Two examples: ½ + ½

### SECTION C

**Q9.** Answer the following about the metals aluminium, copper, iron and zinc: **[3]**

(i) Al > Zn > Fe > Cu [1]

(ii) No. Copper is less reactive than iron, so it cannot displace iron. [1]

(iii) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ [1]

**Q10.** Mercury is obtained from its ore, cinnabar (HgS). Explain the steps with balanced equations. **[3]**

**Answer:** Mercury is low in the reactivity series, so heating alone is enough. Cinnabar is first roasted in air to form mercuric oxide: $2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$. On further heating, mercuric oxide is reduced to mercury: $2HgO \rightarrow 2Hg + O_2$

- Low reactivity, so heating is enough: 1
- Roasting equation: 1
- Reduction equation: 1

**OR, answer:** The reaction of iron(III) oxide with aluminium. It is highly exothermic, so the iron is produced in the molten state: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ + heat. Use: joining railway tracks or cracked machine parts.

- Description: 1
- Balanced equation: 1
- Use: 1

## SECTION D

**Q11.** Answer the following: **[5]**

(i) Roasting: a sulphide ore is heated strongly in excess air, e.g. $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$. Calcination: a carbonate ore is heated strongly in limited air, e.g. $ZnCO_3 \rightarrow ZnO + CO_2$ [3]

- Difference: 1
- Each equation: 1 + 1

(ii) Methods (any two): painting, oiling, greasing, chrome plating, anodising, making alloys. Galvanisation means coating iron or steel with a thin layer of zinc. The zinc protects the iron even if the coating is broken. [2]

- Two methods: 1
- Galvanisation: 1

**OR, answer:** (a) There are strong forces of attraction between the oppositely charged ions, so a large amount of energy is needed to break them. (b) Water is a polar solvent and can separate the ions. Kerosene and petrol are non-polar. (c) In the solid state the ions are held in fixed positions and cannot move. In the molten state the ions are free to move and carry charge.

- (a) 2
- (b) 1
- (c) 2

## SECTION E

**Q12.** Pure gold, known as 24 carat gold, is very soft and is not suitable for making jewellery. It is alloyed with silver or copper to make it hard. In… **[4]**

(i) 22 parts of pure gold alloyed with 2 parts of copper or silver. [1]

(ii) Copper and tin. [1]

(iii) Solder has a low melting point. Other reasons (any one): to increase hardness or strength, to reduce corrosion, to change electrical conductivity. [2]

- Low melting point: 1
- Another reason: 1